

DLA Draft

Developmental Learning Architecture: Learning History, Fast-Weight Traces, and Future Adaptation

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Abstract

A conventional neural learner is treated as a fixed function trained once by an external optimizer. This paper studies a more developmental question: does a learner’s prior learning history change how it will learn in the future, and *which part of the learner* carries that change? We use DLA, an architecture that keeps a standard Transformer backbone unchanged but augments every weight with a fast trace w_{fast} , a per-parameter plasticity P , and sleep-boundary consolidation, so the learner’s own state (not its architecture or its task list) is the object of study. On a 6.59M Chinese character GPT we establish three things. (i) *Where history lives*: replacing a hard→easy learner’s w_{fast} with an easy→hard learner’s significantly degrades future adaptation on an unseen domain ($n=12$, gain@40 difference ≈ -0.019 , CI excludes zero, sign $p=0.019$, 10/12 negative); norm-matched, shuffled and module-localized controls show the effect is not magnitude, randomness, or a single embedding/attention/MLP module. (ii) *What the mechanism is not*: an explicit code audit shows DLA contains no alignment objective and no parameter-level selection — its success signal is a global scalar, and the success-gated eligibility trace Q is numerically ($\sim 0.4\%$ of the write

energy) and causally inert. (iii) *What the mechanism is*: the history effect is carried by (a) the self-organized directional trace that w_{fast} integrates from its Adam-shaped gradients, whose per-seed alignment with the future gradient predicts adaptation ($r=0.87$; survives norm-confound controls and module-localizes to the same MLP/attention modules implicated causally), and (b) the direct fast→slow write at task boundaries, whose **coordinate allocation is functionally necessary**: keeping the write energy identical but shuffling which coordinates receive it removes the effect ($n=12$, paired $t\approx 4.9$), i.e. the system exhibits self-organized, allocation-level selectivity *without any selection objective*. We find no evidence that more experience monotonically accelerates learning (multiple controlled longitudinal null results).

1. Introduction

1.1 What problem exists

Modern neural networks are trained and then frozen. When they learn a second task, they either overwrite the first (catastrophic forgetting) or treat the two tasks as independent (Kirkpatrick et al., 2017; Zenke et al., 2017; De Lange et al., 2021). More fundamentally, standard training does not allow the model itself to become a different kind of learner: the learning algorithm is fixed outside the model, whether it is an external optimizer or a meta-learned rule (Andrychowicz et al., 2016; Miconi et al., 2019; Beaulieu et al., 2020).

We ask a problem that sits before “continual learning” and “meta-learning” — yet is closely tied to a long-standing empirical question about task and curriculum ordering (Poirier & Silver, 2005; Bell & Lawrence, 2022; Li & Hiratani, 2025):

Can what a model has experienced change how it will learn in the future, and which part of the model carries that developmental change?

1.2 Why it matters

If learning history only changes stored knowledge, the learner remains a static function with a growing database. If learning history also changes the learner state, then a small model could, in principle, become a better learner through experience—without changing its architecture. This developmental framing has a deep biological precedent: complementary

learning systems (CLS) theory explains memory/learning trade-offs by the interaction of a fast hippocampal system and a slow neocortical system, consolidated over time (McClelland, McNaughton & O'Reilly, 1995; Kumaran, Hassabis & McClelland, 2016). Recent empirical results further show that plasticity itself is a fragile resource that ordinary training can consume (Dohare et al., 2024; Lyle, Rowland & Dabney, 2022; Nikishin et al., 2022), so “what kind of learner a model is” is not fixed by its architecture alone.

The practical significance is twofold. First, it tells us which parts of a model to preserve or transfer between learning phases. Second, it gives a falsifiable framework for claims such as “more experience makes models learn faster” — a claim we test and do not find support for. Throughout, we treat every mechanistic claim as a diagnosis: we state at which level it is established (causal, controlled, correlational, or absent) rather than over-claiming a mechanism.

2. How existing work addresses the problem

- **Catastrophic forgetting methods** (regularization, replay, EWC, pseudorehearsal) address memory, not the change of the learner itself (Kirkpatrick et al., 2017; Zenke et al., 2017; Robins, 1995; De Lange et al., 2021).
- **Fast-weight models** add a temporary memory but usually keep the learning rule external. The idea goes back to fast-weight memories (Hinton & Plaut, 1987; Ba et al., 2016) and has seen a modern revival: linearised attention is formally equivalent to fast-weight programming (Schlag, Irie & Schmidhuber, 2021), in-context learning can be understood as implicit gradient descent / fast-weight updates inside a Transformer (von Oswald et al., 2023), and recent sequence models introduce trainable “neural memory” modules with fast updates alongside slow base weights (Behrouz, Zhong & Mirrokni, 2025). Fast/slow dual-module designs also appear in RLHF pipelines (“fast-slow chasing” in online DPO; Qi et al., 2024). In most of these, the fast store is either a memory mechanism or an engineering device; DLA instead treats fast/slow as a *developmental state* whose history we causally dissect.
- **Complementary learning systems and consolidation.** CLS theory attributes the complementary benefits of fast, instance-based and slow, generalising stores to their interaction and to offline consolidation (McClelland et al., 1995; Kumaran et al., 2016). Machine-learning instantiations usually realise this as experience replay; DLA instantiates the fast/slow split as an internal, per-

parameter weight state with sleep–boundary consolidation and no external replay buffer.

- **Meta-learning** optimizes a learning rule over many episodes; it asks whether a rule can be learned, but rarely whether one individual’s rule develops within a single lifetime (Andrychowicz et al., 2016; Beaulieu et al., 2020).
 - **Learned plasticity** (differentiable plasticity, ANML) shows that plasticity can be meta-learned, but does not separate *body state* from *learning-rule parameters* as causal carriers (Miconi et al., 2019; Beaulieu et al., 2020). Complementary findings that plasticity degrades over long training—capacity loss and primacy bias—motivate making plasticity itself a state to maintain (Dohare et al., 2024; Lyle et al., 2022; Nikishin et al., 2022).
 - **Task and curriculum ordering.** The order in which tasks are learned measurably changes continual-learning outcomes and the amount of forgetting (Bell & Lawrence, 2022; Li & Hiratani, 2025), affects the consolidation of task knowledge (Poirier & Silver, 2005), and is the core object of curriculum learning (Wang, Chen & Zhu, 2022). This literature mostly measures order effects on *stored knowledge* about seen tasks. We ask a complementary question: does order change the *learner state* itself, such that future adaptation to a *never-seen* task changes—and which state component carries that effect?
 - **We therefore design DLA** to ask a mechanistic question that is usually skipped: given the same task sequence, which state component carries history-dependent future adaptation?
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3. Our solution: DLA

3.1 Design principle

We do not modify the Transformer skeleton. Instead, each weight matrix is accompanied by:

- W_{slow} : long-term consolidated knowledge (the base weights)
- W_{fast} : a fast learning trace
- P : per-parameter plasticity
- Q : a sleep eligibility trace

Effective weights:

$$W_{\text{eff}} = W_{\text{slow}} + \text{softplus}(P) * W_{\text{fast}}$$

Structurally, this additive form resembles parameter-efficient transfer methods that freeze a base network and attach small trainable additive weights—adapters (Houlsby et al., 2019), LoRA (Hu et al., 2021), and the unified additive-PEFT view (He et al., 2022). DLA differs in three ways that matter for development: (i) the additive pathway is a *within-lifetime developmental state* updated online by the learner’s own wake/sleep dynamics, not a separately fine-tuned module; (ii) it is shared across tasks and persists (subject to decay and consolidation) rather than being re-initialised per task; and (iii) its contribution is gated per parameter by a learned plasticity $\text{softplus}(P)$. The fast/slow decomposition follows complementary learning systems theory (McClelland et al., 1995; Kumaran et al., 2016).

3.2 Wake and sleep (exact update rules used in this paper)

Wake step (per batch, per parameter matrix):

```

g          = dL/dW_eff          (backprop)
m <- b1 m + (1-b1) g ; v <- b2 v + (1-b2) g*g      (Adam moments)
adam       = m_hat / (sqrt(v_hat) + eps)
dw         = -eta_fast * softplus(P) * adam - fast_decay * W_fast
W_fast    += dw
dp         = eta_plast * progress * relevance - stability*(P - P0)
success    = clamp((loss_ema - loss)/(loss_ema+1e-4), 0, 1)  # GLOBAL scalar
Q          = (1 - alpha_q) Q + alpha_q * (dw * success)

```

success is one scalar shared by every parameter: the only “selection signal” in the code is step-level and global; there is no per-parameter or per-subspace term in Q .

Sleep (task boundary) — two write pathways into W_{slow} :

```

W_slow += beta * Q + gamma * W_fast      # Q pathway + DIRECT fast->slow
W_fast *= consolidate_fast_decay
Q      *= consolidate_q_decay
reset Adam moments

```

with $\beta \approx 1.0$, $\gamma \approx 0.15$, decay 0.5/0.7 (config values used at probe time; probes run with default rule parameters). The second term is an explicit **unselected bypass**: W_{fast} is copied into W_{slow} at a uniform coefficient regardless of any gate.

Algorithm 1: DLA wake-sleep cycle

```

1: for task t in curriculum do
2:   for step in 1..T do

```

```

3:     compute g = dL/dW_eff
4:     update W_fast with Adam(g), gated by softplus(P)
5:     update P from progress/relevance; Q += dw * success (global
    scalar)
6:   end for
7:   consolidate: W_slow += beta*Q + gamma*W_fast
8:   decay: W_fast *= c, Q *= d ; reset moments
9: end for

```

3.3 Experimental design to isolate development

To test whether history changes the future learner, we: 1. Build two histories with the same three domains but different order: easy→hard (EH) and hard→easy (HE). 2. Probe each individual on a never-seen domain D. 3. Perform body/component cross-injection to see which state component transfers the history effect. 4. Run norm-matched, shuffled, and module-localized controls to rule out magnitude/randomness. 5. Compare with standard baselines (AdamW, replay, EWC). 6. Run consolidation-pathway ablations (Q-only / direct-only / none) and an energy-matched **allocation shuffle** to ask whether consolidation—and, if so, *where* it writes—matters. 7. Audit the code statically for the presence of explicit alignment or selection objectives.

4. Experimental setup

Primary backbone: 6.59M Chinese character GPT (6 layers, 8 heads, 256 dim, vocab 7280), pretrained on Chinese Wikipedia. Curriculum domains: Wikipedia, SFT-style QA, science Wikipedia; unseen domain D is a science slice. Second setting: ~10.65M Shakespeare character GPT (vocab 65). Metrics: gain@40, LE_D, T80, paired t / bootstrap CI, sign test. All within-seed, within-arm contrasts are paired; raw EH/HE differences are reported with their own variance and are not assumed stable across seeds.

Table 1: Common experimental hyperparameters.

Hyperparameter	Value
Block size	128
Batch size	32
Steps per task	40
AdamW learning rate (baselines/DLA base)	1e−4

Hyperparameter	Value
EWC lambda	1e3
Replay buffer capacity	48 batches
Seeds (primary setting, causal/ablation)	12 (0–11)
Seeds (baselines)	5
Seeds (second setting)	2
Consolidation coefficients (beta, gamma, decays)	$\approx 1.0, 0.15, 0.5, 0.7$

5. Results: how well does it work

5.1 Fast/slow separation protects consolidated memory

DLA matches tuned AdamW on new-domain adaptation (gain +14.0% vs +13.0%) while slow-memory forgetting is ≈ 0 (−0.4%). The mechanism does not sacrifice memory for plasticity.

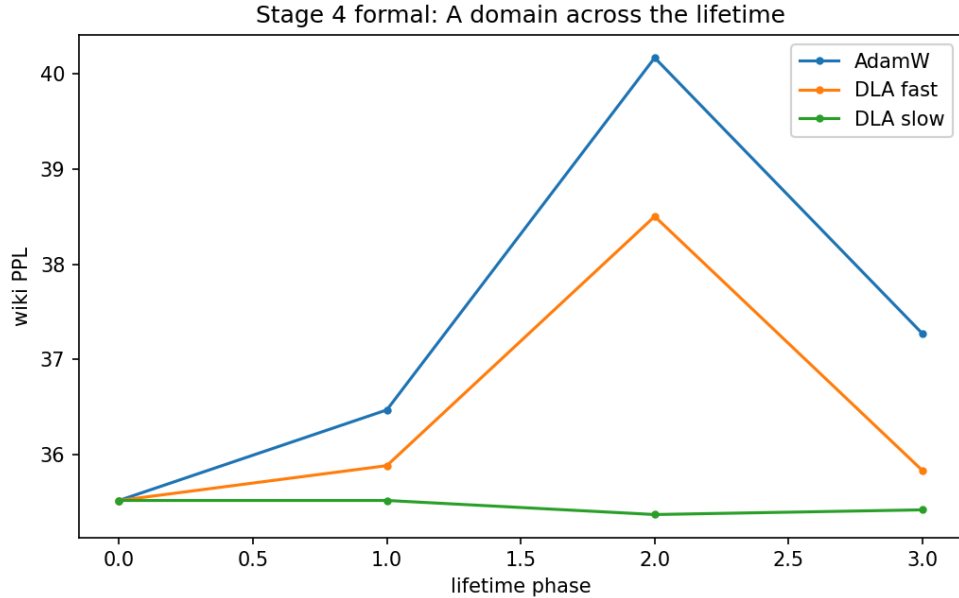


Figure 1a: Fast/slow separation and retention across domains.

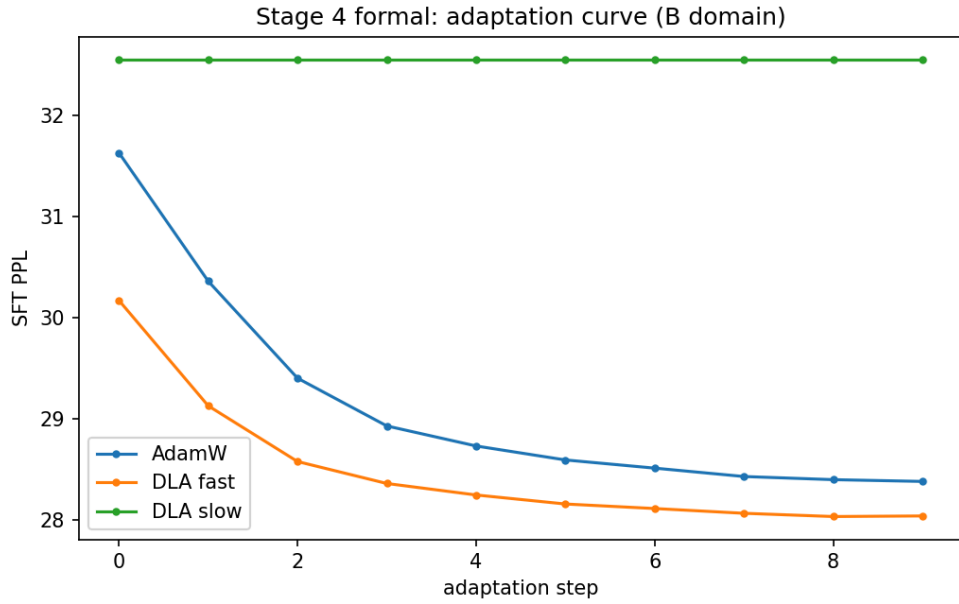


Figure 1b: Adaptation curve on the new domain.

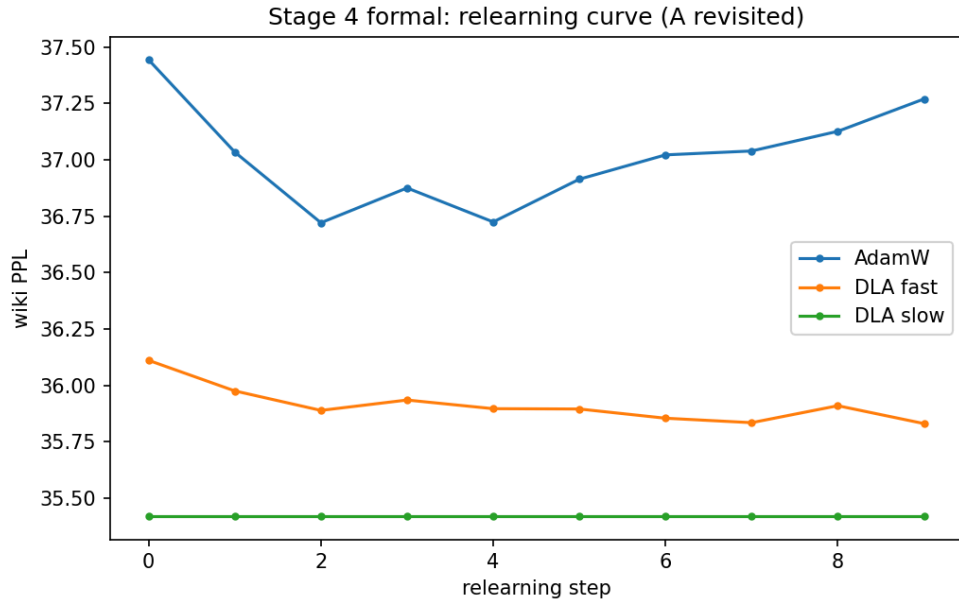


Figure 1c: Relearning curve.

5.2 Learning history changes future adaptation

Across histories, HE learners adapt better to unseen D than EH learners on average, mirroring continual-learning and curriculum order effects measured on seen tasks (Bell & Lawrence, 2022; Li & Hiratani, 2025; Poirier & Silver, 2005). The distinctive feature is that D was never seen in either history, so the order effect must act through the learner’s internal state rather than stored content about D. The raw EH/HE gap is **not stable across seeds** ($n=12$ paired HE–EH $dz \approx 0.6$; several seeds reverse), so every strong claim below uses *within-seed*, *within-arm* contrasts (swap, ablation, shuffle), not the raw gap.

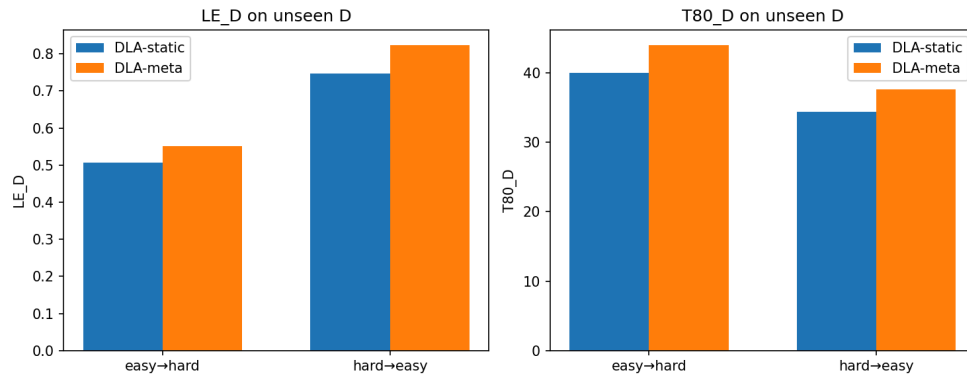


Figure 2: History effect on unseen D (2x2 development experiment).

5.3 The effect is not carried by the learning-rule parameters ϕ

Body $\times\phi$ cross-injection shows that swapping ϕ between histories changes future adaptation much less than swapping body state. ϕ is not a stable/dominant carrier in this setting.

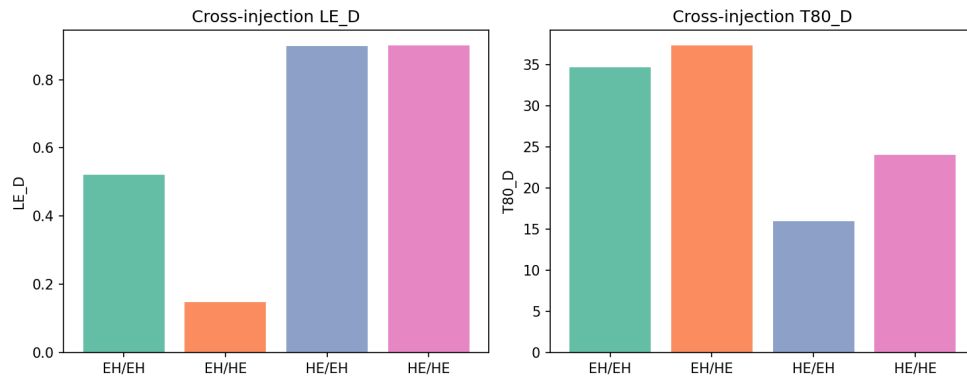


Figure 3: Body $\times \phi$ causal dissection.

5.4 W_{fast} is a causal component (carrier localization)

Core result (n=12):

condition	mean gain@40
HE/HE	+0.014
HE + EH W_{fast}	-0.005

Paired contrast: mean -0.019 , 95% CI $[-0.033, -0.005]$, Cohen's $d \approx -0.7$, sign $p=0.019$ (10/12 negative). Initial PPL differences are small; the effect appears in the adaptation trajectory.

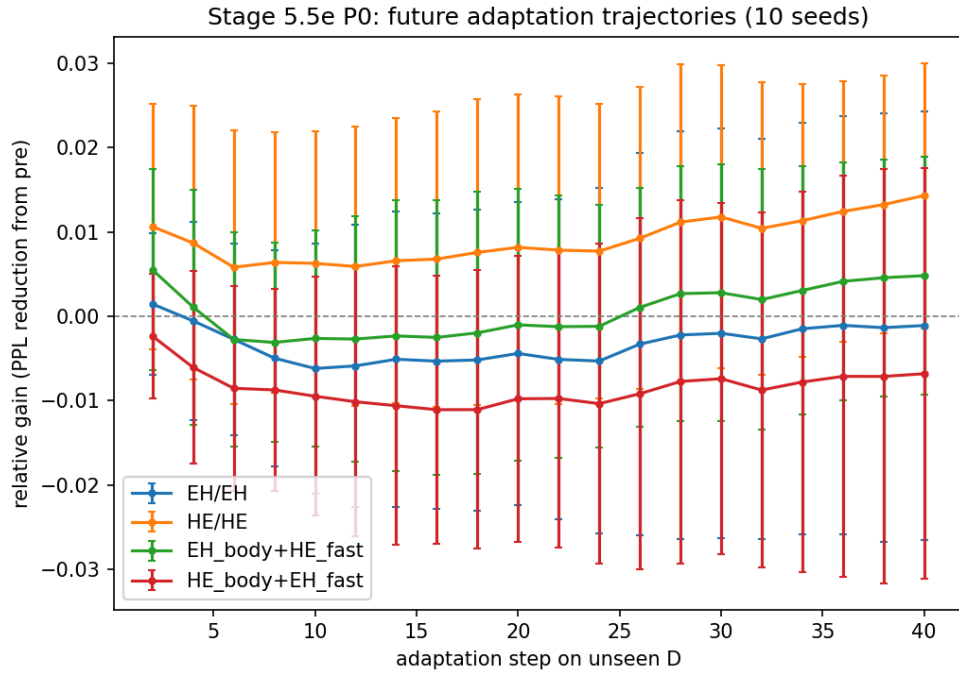


Figure 4: P0 future-adaptation trajectories.

5.5 Controls: magnitude, randomness, module

control	mean Δ gain@40	95% CI	d
raw EH W_fast	-0.019	[-0.034, -0.005]	-0.71
norm-matched EH W_fast	-0.022	[-0.034, -0.009]	-0.90
shuffled EH W_fast	-0.017	[-0.023, -0.009]	-1.25
EH embedding W_fast	-0.006	[-0.009, -0.002]	-0.87
EH attention W_fast	-0.012	[-0.018, -0.006]	-1.06
EH MLP W_fast	-0.017	[-0.025, -0.008]	-1.05

Norm-matching does not rescue the effect; shuffling also does not remove it. Module localization is suggestive of MLP and attention, with embedding weaker.

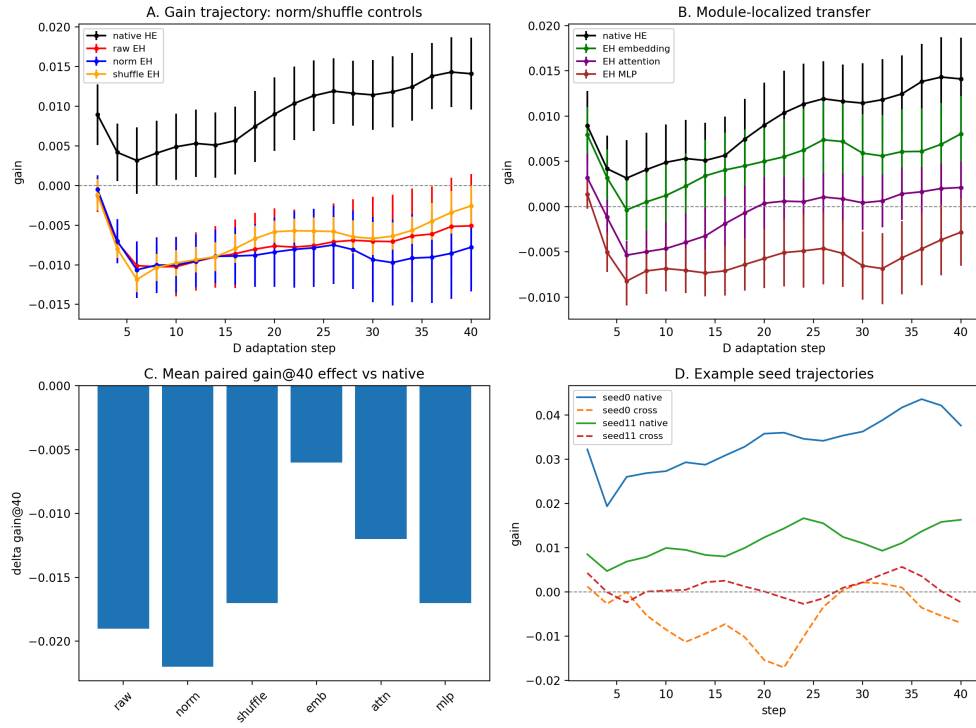


Figure 5: W_{fast} cross-injection and full future-adaptation trajectories. A: norm/shuffle controls. B: module-localized transfer. C: paired gain@40 effects. D: example trajectories.

5.6 Fair baselines

method	EH	HE	HE-EH
AdamW	0.339	0.675	+0.336
AdamW+replay	0.375	0.961	+0.586
EWC	0.292	0.895	+0.603
DLA	0.517	0.709	+0.192

DLA is the most robust after an easy $\rightarrow$ hard history; standard learners show larger HE performance but larger EH degradation.

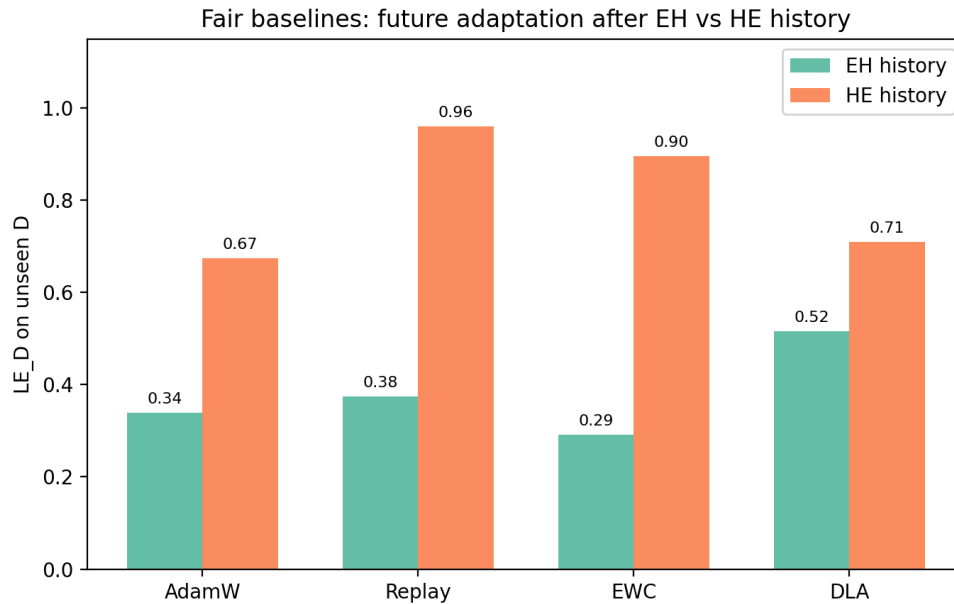


Figure 6: Fair baselines — future adaptation after EH vs HE history.

5.7 Second-setting replication

Shakespeare char GPT, 2 seeds: HE/HE gain@40 +0.052/+0.042; HE+EH fast +0.004/−0.001. The destructive effect replicates in direction (n=2, exploratory).

5.8 Negative result: more experience does not necessarily make learning faster

Stage 5 (difficulty-normalized): flat LE. Stage 6 (matched difficulty): p=0.17. Stage 7 (cross-domain, 20 seeds): p=0.82. Stage 8 (physics near-transfer, 20 seeds): d=−0.17. We do not claim “more experience → faster learning”. This null is consistent with a growing plasticity literature (Nikishin et al., 2022; Lyle et al., 2022; Dohare et al., 2024).

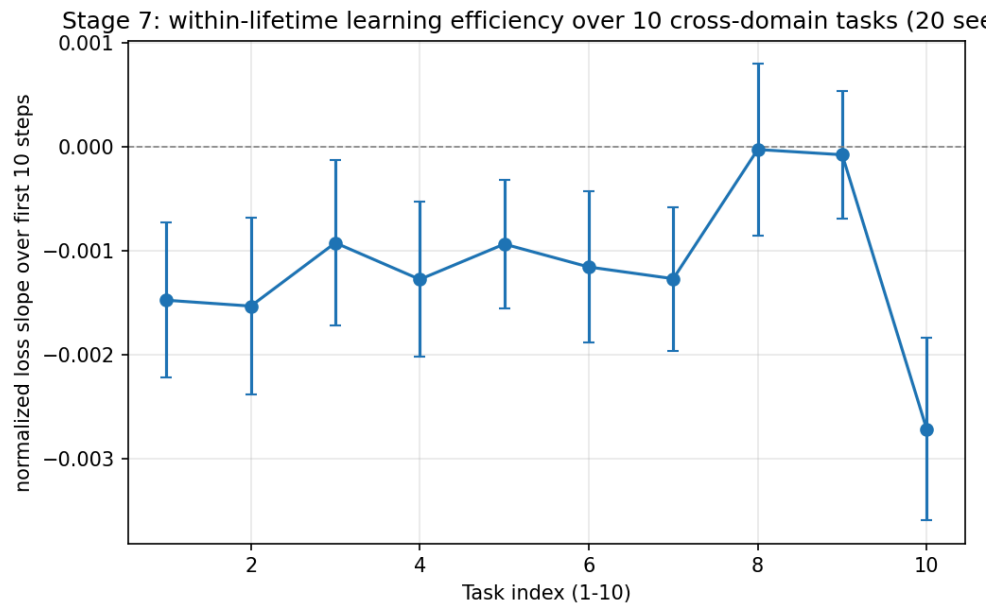


Figure 7a: Negative result — cross-domain longitudinal (Stage 7, 20 seeds).

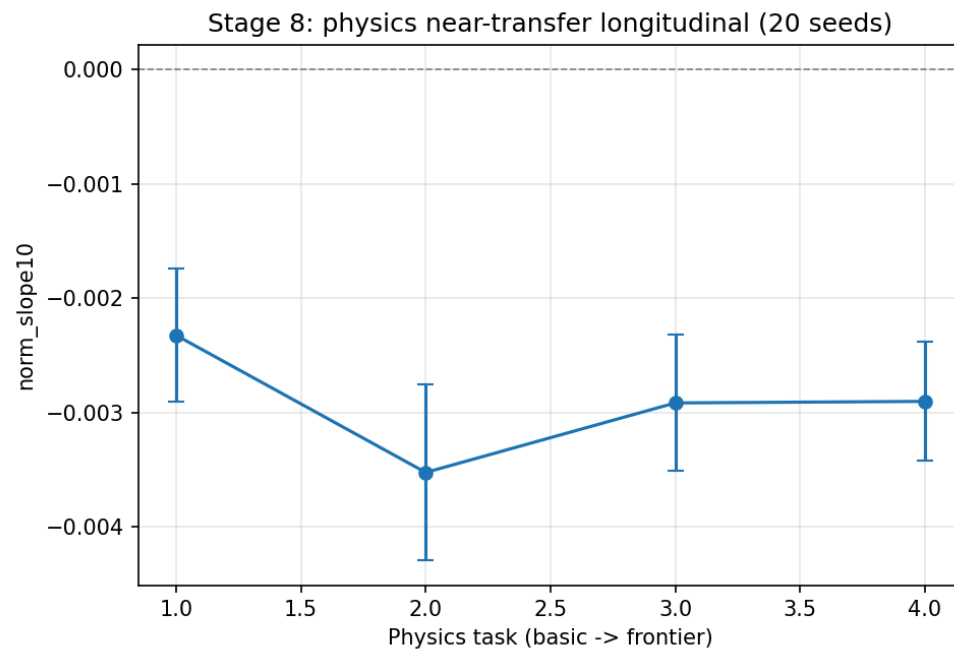


Figure 7b: Negative result — physics near-transfer (Stage 8, 20 seeds).

6. Mechanism audit: what the effect is, and what it is not

6.1 There is no explicit alignment or selection mechanism in the code

A static audit of the full wake/sleep/meta path (`dla/transformer_dla.py: dla_step, dla_sleep, meta_unroll_loss`) shows: – **No alignment objective.** No term compares a gradient to a history direction; no cosine/normalized-dot exists anywhere in the Transformer-DLA path. – **No parameter-level selection.** success is a single scalar from EMA losses (same coefficient for every parameter); $Q = \text{EMA}(\text{dw} \cdot \text{success})$ is therefore a *step-level global gating*, not parameter- or subspace-level selection. The only per-parameter signal (relevance) feeds P, a positive coordinate rescale, never Q. – The directional content that exists is **emergent**: W_{fast} is a leaky integrator (decay 0.02/step $\Rightarrow \approx 50$ -step horizon $\approx$ one stage) of the Adam-shaped gradient – $\text{eta} \cdot \text{softplus}(P) \cdot \text{adam}$, so the “direction of history” is an accumulated, normalised recent-gradient trajectory — not an imposed target.

6.2 The directional character is measurable, module-aligned, and not a norm artifact (correlational)

From 48 deterministic replays of the stored D-probe and the 24 saved bodies (seeds 0–11): – The per-seed alignment $\cos(g_0(\text{HE}), \Delta W_{\text{fast}})$ of the initial D-gradient with the same seed’s history contrast predicts HE adaptation: $r=0.87$, permutation $p \approx 0.000$, leave-one-seed-out $r \in [0.81, 0.91]$, and it is the only effect in the correlation table surviving Benjamini–Hochberg FDR ($q \approx 0.003$). – Controlling for $\|\Delta W\|$, relative norm, $\cos(\text{EH}, \text{HE})$, and gradient norm leaves partial $r \approx 0.77$ — not a simple magnitude confound. True same-seed pairing ($r \approx 0.86$) beats cross-seed pairing (0.66) and shuffled Δ (0.30). – Module alignment mirrors the causal module controls: MLP $r \approx 0.87$, attention $r \approx 0.83$, embedding $r \approx 0.68$. – **Macroscopic geometry is not the carrier**: $\|\Delta W_{\text{fast}}\|$ and $\cos(\text{EH}, \text{HE})$ are indistinguishable from same-history cross-seed noise (3.34 vs 3.21–3.26; 0.579 vs 0.60–0.61), and PCA shows no dominant shared direction (PC1 $\approx 20\%$ variance). – *Caveat*: this whole block is correlational; we do not claim directional alignment is causal (no direction manipulation in this paper).

6.3 Consolidation acts through the direct fast→slow write, not through Q

Measured on the saved bodies: mean $\|Q\| \approx 0.002$ vs $\|W_{\text{fast}}\| \approx 3.6$; estimated $Q \rightarrow W_{\text{slow}}$ write is $\approx 0.4\%$ of the direct write per sleep. During the D-probe run_transfer never calls sleep, so Q cannot act on the measured outcome except via earlier W_{slow} writes.

Consolidation ablation (n=12, HE arm, gain@40):

variant	mean gain@40	paired t vs full	paired t vs direct
full (archive)	+0.0143	—	—
direct ($\gamma \cdot W_{\text{fast}}$ only)	+0.0148	+0.59	—
nocons (no write)	+0.0036	-5.53	+5.30
qonly ($\beta \cdot Q$ only, n=4)	+0.0094	—	$\approx$ nocons

direct $\approx$ full; removing consolidation (nocons) significantly drops the HE arm; Q-only (n=4) behaves like no-consolidation $\Rightarrow$ the success-gated Q channel has no independent causal contribution.

6.4 The coordinate allocation of the direct write is necessary (self-organized selective allocation)

Energy-matched control: shufwrite keeps every per-sleep write energy identical to direct but randomly permutes, within each matrix, which coordinates receive the increment (only the allocation is destroyed).

HE arm, n=12: shufwrite mean +0.0042 vs direct +0.0148 $\rightarrow$ paired $t=4.86$; shufwrite $\approx$ nocons ($t=-0.38$) $\ll$ direct ($t=4.86$).

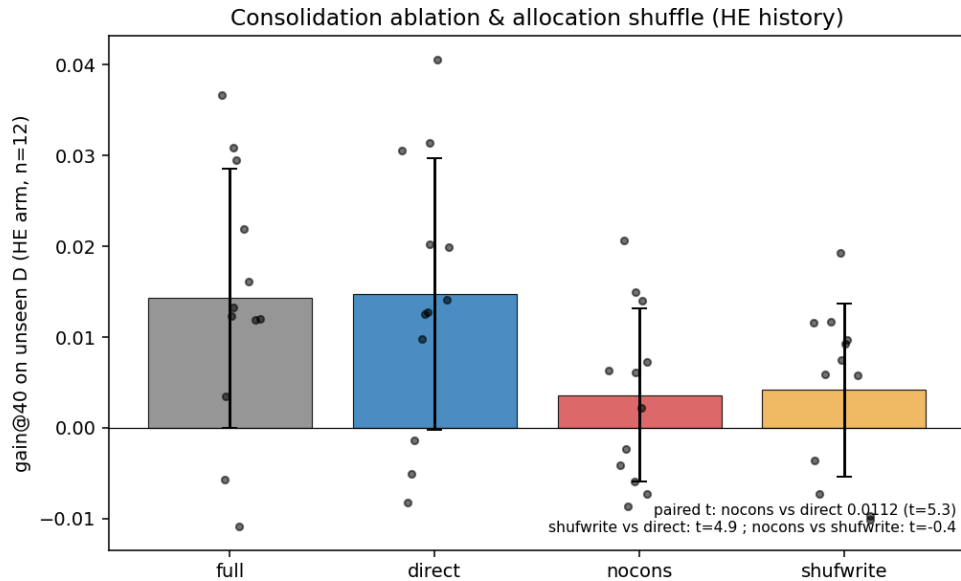


Figure 8: Consolidation ablation & allocation shuffle (HE arm, n=12).

Because the increment is just $\gamma \cdot W_{\text{fast}}$ with a uniform scalar coefficient, “where it writes” is entirely determined by the self-organized structure of W_{fast} . Destroying that structure with an energy-matched shuffle removes the benefit $\Rightarrow$ the system shows **selectivity of allocation without any selection objective**: a form of emergent, allocation-level selective consolidation.

6.5 Statement-level summary

Claim	Implemented in code?	Observed?	Causal evidence?	Status
History effect localized to W_{fast} (carrier)	—	yes	strong (swap, n=12, controls, 2nd setting)	established
Directional alignment as explicit operator/loss	no	—	—	absent
Emergent directional character (predictive, module-aligned)	—	yes (r=0.87)	correlational only	supported as <i>character</i> , not mechanism

Claim	Implemented in code?	Observed?	Causal evidence?	Status
Selective consolidation via success-gated Q	step-level scalar only	$Q \approx 0.002$; qonly $\approx$ nocons	inert	refuted in this setting
Direct fast $\rightarrow$ slow write carries effect	yes ($\gamma \cdot W_{\text{fast}}$)	yes	strong (n=12, $t \approx 5.3$)	established
Allocation of the direct write is necessary	emergent from W_{fast}	yes	strong (energy-matched shuffle, $t \approx 4.9$)	established
More experience $\rightarrow$ faster learning	—	null (stages 5–8)	—	not supported

7. Limitations

- Second-setting $n=2$; baselines $n=5$; all causal/ablation claims are from one 6.59M Chinese character GPT.
- Directional-alignment evidence is correlational (no direction manipulation).
- Retention (protection of earlier-task memory) is not measured in the same protocol as future adaptation (probes never sleep); the two halves of consolidation are separate in this paper.
- The raw EH/HE adaptation gap is seed-noisy; strong claims rest on within-seed paired contrasts.
- No pre-registration; we report $n=10 \rightarrow n=12$ and every null transparently.
- Sleep consolidation writes only a small amount into w_{slow} per boundary; the effect we attribute to the direct write is therefore about *where* the (small) write lands, not its bulk.

8. Conclusion and significance

Without any explicit alignment or selection objective, a learner’s history still leaves a persistent, causal trace in its fast weights: replacing that trace changes how the learner will adapt to something it has never seen. Mechanistically, the trace is a self-organized directional structure (w_{fast} as a leaky integrator of its Adam-shaped gradients) whose per-seed alignment with the future task predicts adaptation and localizes to the same modules that are causally implicated; at task boundaries, the fast→slow direct write — not the success-gated eligibility trace — transfers part of that structure into the slow store, and *which coordinates* receive the write is functionally necessary (energy-matched shuffle control). The success-gated “selective consolidation” the architecture nominally contains is numerically and causally inert. We therefore reframe the contribution: not “fast/slow weights as a new mechanism”, but a controlled, honest decomposition of *where* developmental change lives in a learner, *what* it consists of (emergent directional structure + allocation-selective writeback), and *what it does not consist of* (explicit alignment, parameter-level selection, or universally faster learning).

Reproducibility

Code: <https://github.com/JayCRL/DLA> Experiment details: docs/experiments.md, paper/overnight_report.md, paper/validation_audit.md. Scripts: stage4_formal.py, stage55e_wfast_p0.py, validation_p0_controls.py, validation_baselines.py, validation_second_setting.py, stage6_longitudinal.py, stage7_cross_domain.py, stage8_physics.py; mechanism audit: analysis/wfast_geom/ (geom.py, replay.py, p1*.py, p2.py, p4_consolidation_geom.py, audit_cheap.py, audit_b3.py, reports under analysis/wfast_geom/report_output/). Per-seed data, scripts and commit hashes for every claim are recorded in the repository; the mechanism-audit runs used a parity-validated harness on an Apple M2 (CPU), identical protocol to the archived server runs (mean $|\Delta \text{ppl}| \approx 0.24$ vs archive).

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